

# Algorithm and Flowchart Explained

## With Examples



The is taken from a Post by Allison Lynch | 11/08/2019

(<https://www.edrawsoft.com/explain-algorithm-flowchart.html>)

Algorithm and flowchart are two types of tools to explain the process of a program. This article extends the differences between an algorithm and a flowchart, and how to create a flowchart to explain an algorithm in a visual way.

Algorithms and flowcharts are two different tools used for creating new programs, especially in computer programming. An algorithm is a step-by-step analysis of the process, while a flowchart explains the steps of a program in a graphical way.

### What is an Algorithm

An algorithm is a step-by-step demonstration of data processing or problem solving. This page will introduce the definition, types and applications of algorithm.

An algorithm can be described as a procedure or formula for problem solving. Algorithms can be widely used in various areas, computer programming, mathematics and daily lives. Then what is the definition of algorithm? How many types are there and in what ways can they be applied?

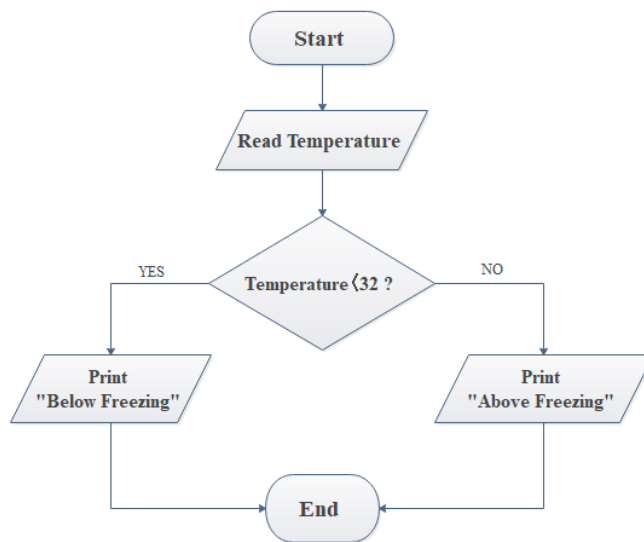
### What is the difference between an Algorithm and Pseudocode

An algorithm is a systematic logical approach used to solve problems in a computer while Pseudocode is the statement in plain English which may be translated later into a programming language (program). An algorithm is the semantic while the pseudo-code is just a syntax of the communication about solving a problem.

An algorithm is a step by step procedure to solve a given problem while a pseudocode is a method of writing an algorithm. Therefore, the pseudocode is the algorithm. Programmers can use informal simple language to write a pseudocode and there is no strict syntax to follow.

### What is a Flowchart?

A flowchart is a diagrammatic representation that illustrates the sequence of operations to be performed to get the solution to a problem. It can be seen from the definition that a flow always accompanies with business or transaction. Not all of the flows, however, are appropriate to be expressed by flowcharts, unless these flows are based on some fixed routines and stable links. Here is an example of the algorithm flowchart.



## Flowchart symbols

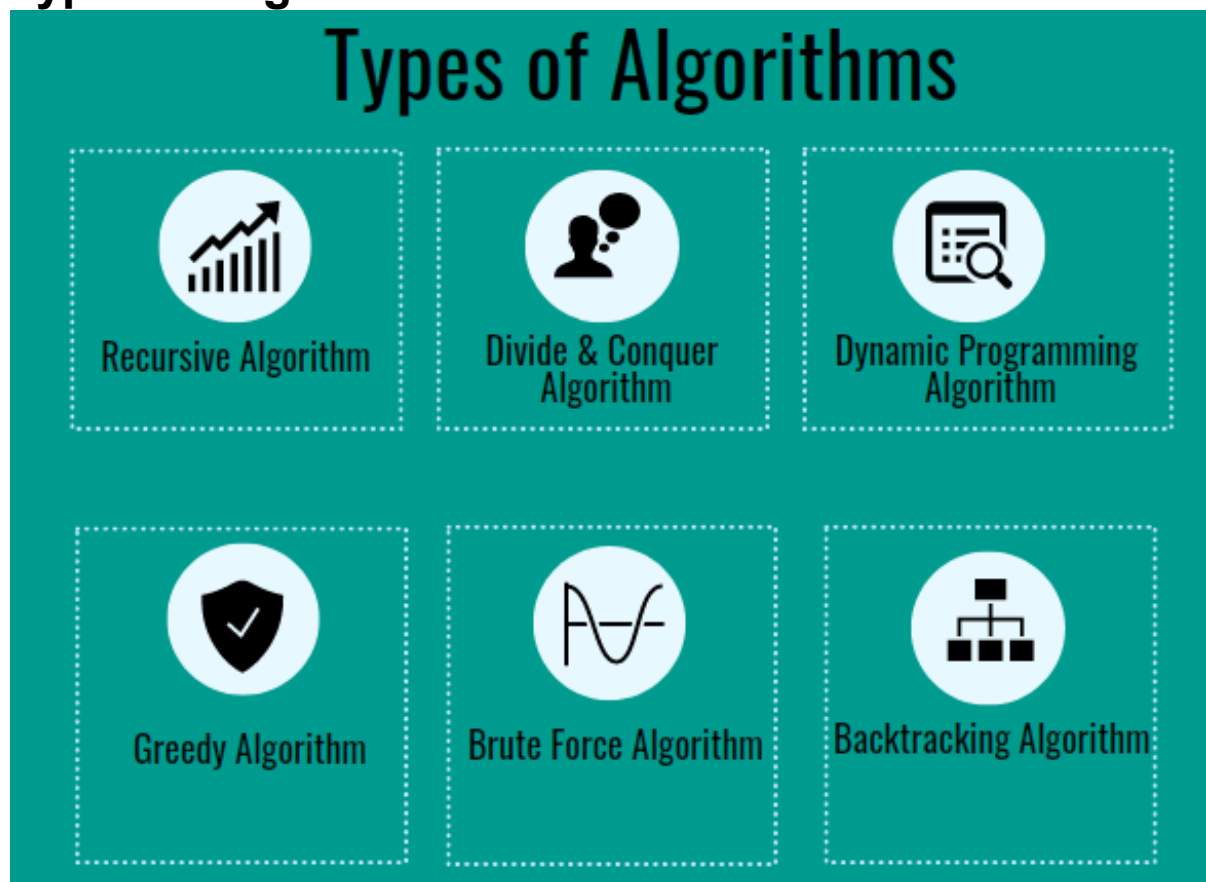
| Name                     | Flowchart Symbol                                                                    | Usage                                                                                                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Process</b>           |   | It is used to represent an event which is controlled within the process. Typically this will be a step or action which is taken. In most flowcharts this will be the most frequently used symbol. |
| <b>Decision</b>          |  | It is used to represent a decision point in the process. Typically, the statement in the symbol will require a 'yes' or 'no' response and branch to different parts of the flowchart accordingly. |
| <b>Terminal Points</b>   |  | It indicates the ending points of a process.                                                                                                                                                      |
| <b>On-page Reference</b> |  | It is used to represent a point at which the flowchart connects with another process. The name or reference for the other process should appear within the symbol.                                |
| <b>Document</b>          |  | It represents a step that results in a document.                                                                                                                                                  |

# Difference between Algorithm and Flowchart

If you compare a flowchart to a movie, then an algorithm is the story of that movie. In other words, an algorithm is the core of a flowchart. Actually, in the field of computer programming, there are many differences between algorithm and flowchart regarding various aspects, such as the accuracy, the way they display, and the way people feel about them. Below is a table illustrating the differences between them in detail.

| Algorithm                                         | Flowchart                                                   |
|---------------------------------------------------|-------------------------------------------------------------|
| It is a procedure for solving problems.           | It is a graphic representation of a process.                |
| The process is shown in step-by-step instruction. | The process is shown in block-by-block information diagram. |
| It is complex and difficult to understand.        | It is intuitive and easy to understand.                     |
| It is convenient to debug errors.                 | It is hard to debug errors.                                 |
| The solution is showcased in natural language.    | The solution is showcased in pictorial format.              |
| It is somewhat easier to solve complex problem.   | It is hard to solve complex problem.                        |
| It costs more time to create an algorithm.        | It costs less time to create a flowchart.                   |

## Types of Algorithm<sup>1</sup>



<sup>1</sup> (Further research can be done to look up examples for each of these types of algorithms)

# How to Use Flowcharts to Represent Algorithms

Now that we have the definitions of algorithm and flowchart, how do we use a flowchart to represent an algorithm?

Algorithms are mainly used for mathematical and computer programs, whilst flowcharts can be used to describe all sorts of processes: business, educational, personal and of course algorithms. Flowcharts are therefore often used as a program planning tool to visually organize the step-by-step process of a program. Here are some examples:

## Example 1: Print 1 to 20:

### Algorithm:

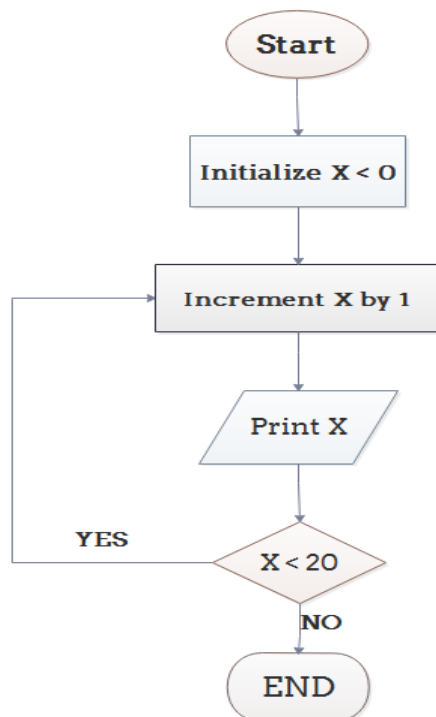
Step 1: Initialize X as 0,

Step 2: Increment X by 1,

Step 3: Print X,

Step 4: If X is less than 20 then go back to step 2.

### Flowchart:



## Example 2: Convert Temperature from Fahrenheit (°F) to Celsius (°C)

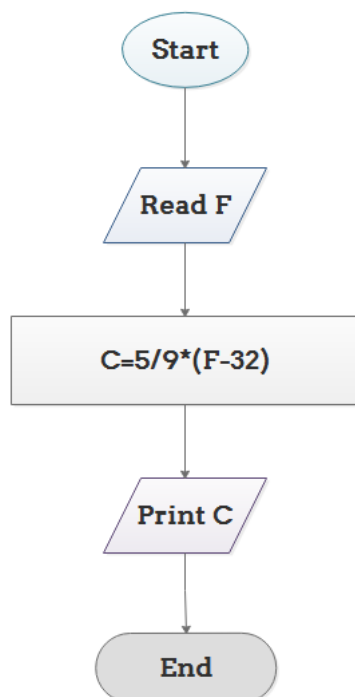
### Algorithm:

Step 1: Read temperature in Fahrenheit,

Step 2: Calculate temperature with formula  $C = 5/9 * (F - 32)$ ,

Step 3: Print C,

### Flowchart:



## Example 3: Determine Whether A Student Passed the Exam or Not:

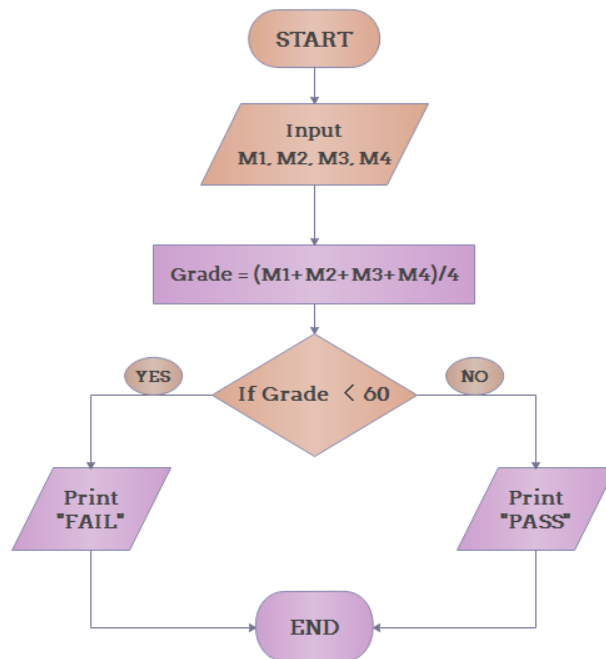
### Algorithm:

Step 1: Input grades of 4 courses M1, M2, M3 and M4,

Step 2: Calculate the average grade with formula "Grade=(M1+M2+M3+M4)/4"

Step 3: If the average grade is less than 60, print "FAIL", else print "PASS".

Flowchart:



## More Flowchart Examples

Posted by Guillaume Lesoin | 02/02/2020

<https://www.edrawsoft.com/flowchart-examples.html>

### Example 1

Draw a flowchart to find the sum of the first 50 natural numbers.

Answer: The required flowchart is given in Fig. 1.

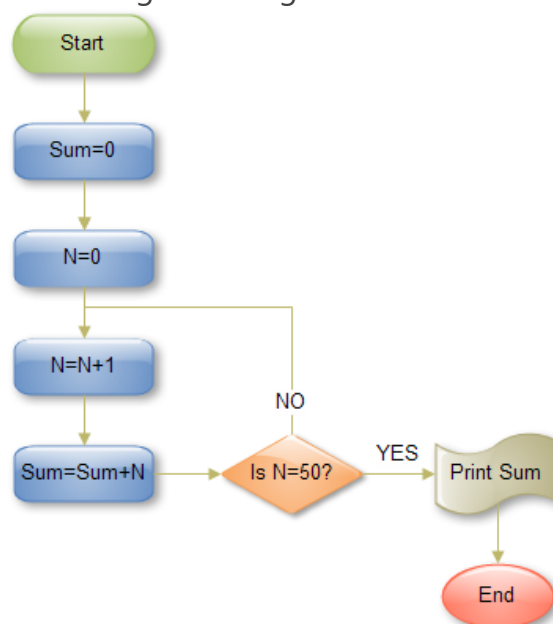


Figure 1

### Example 2

Draw a flowchart to find the largest of three numbers A, B, and C.

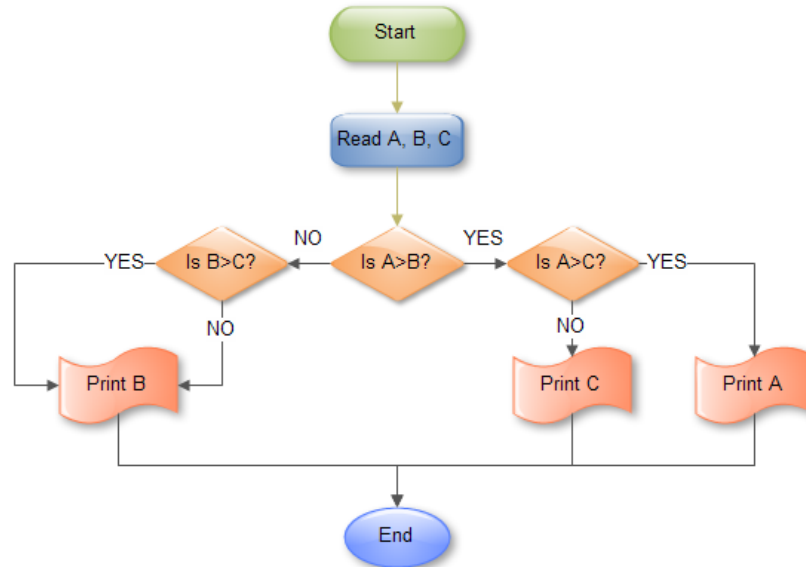


Figure 2

### Example 3

Draw a flowchart for computing factorial N ( $N!$ )

Where  $N! = 1 \times 2 \times 3 \times \dots \times N$ .

The required flowchart has been shown in fig 3

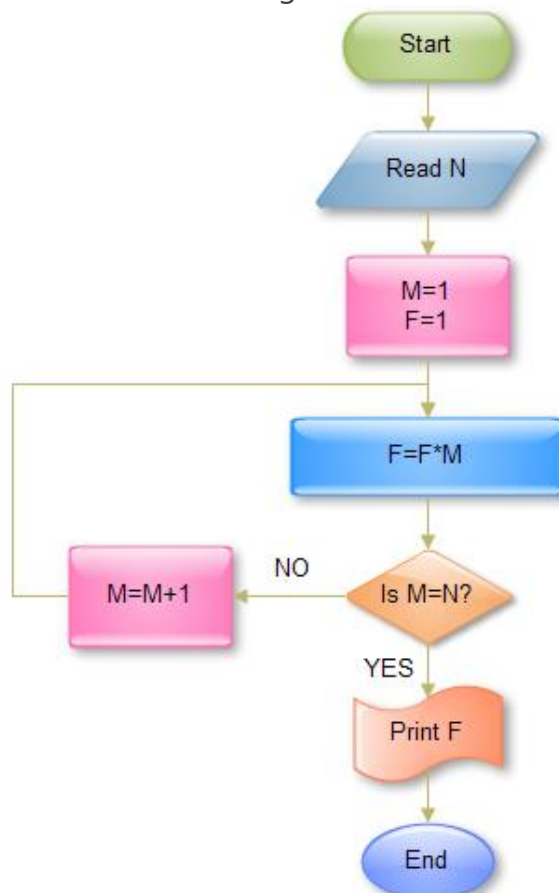


Figure 3

## Example 4

A product assembly team in a gaming machine manufacturer were looking for ways of building the products more efficiently. They broke down the assembly process into a set of Flowcharts, showing how sub-assemblies were made and then built into the final product. Analysis of the reel assembly process revealed two improvements:

1. The kit of parts was already checked by the kit assembly line, who were sometimes careless, as they knew the kit would be rechecked. The assembly line process was improved so the check here could be removed. This saved over two minutes per reel in checking, and up to fifteen minutes when the kit was faulty.
2. Fitting the reel band after the reel had been attached to the base was awkward. Fitting the band before the reel was attached to the base was more comfortable and saved about a minute per reel.

The process Flowcharts, before and after improvement, are shown in Fig. 4.

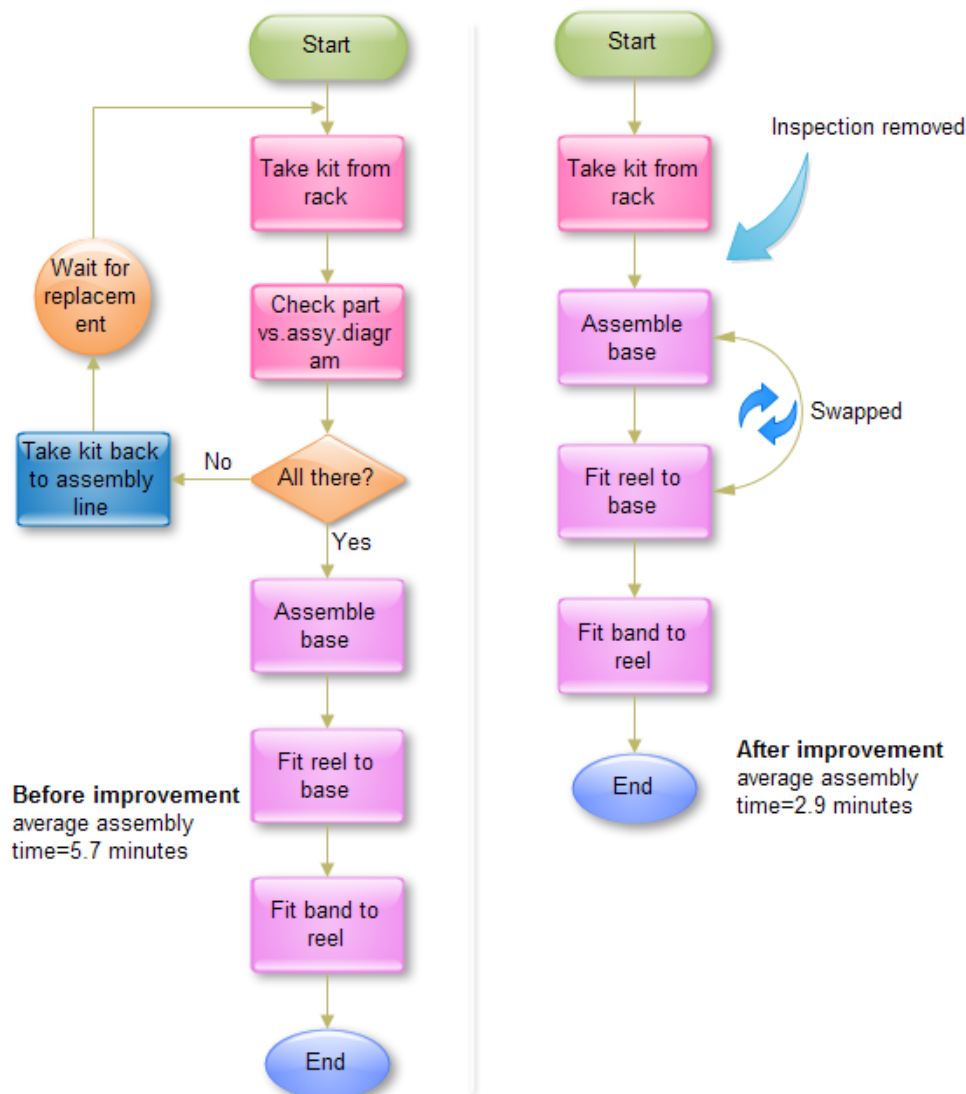


Figure 4



## Conclusion

From the above we can come to a conclusion that a flowchart is pictorial representation of an algorithm, an algorithm can be expressed and analyzed through a flowchart.

An algorithm shows you every step of reaching the final solution, while a flowchart shows you how to carry out the process by connecting each step. An algorithm uses mainly words to describe the steps while a [flowchart uses the help of symbols](#), shapes and arrows to make the process more logical.

## References

<https://www.edrawsoft.com/explain-algorithm-flowchart.html>

<https://www.edrawsoft.com/algorithm-definition.html>

## Other readings:

[Flowchart Tutorials - Flowchart Resource Center](#)